Exercises: Concentration Inequalities, LLN, CLT & Convergence Mathematical Foundations for Machine Learning

$P(X \geq a) = E[\mathbf{1}_{X \geq a}]$ -> $P(X \geq a) \leq \frac{E[h(X)]}{h(a)}$

A random variable will not be many times much bigger than it's expected value -> else it's expected value would be bigger

Concentration Inequalities

Exercise 1.1 [Easy] $Z = |X - \mu|$ abs. deviance from mean $h(z) = z^2 \mathbf{1}_{z \geq 0}$ -> $P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$

Let X be a random variable with $\mathbb{E}[X] = 5$ and $\mathrm{Var}(X) = 4$.

a) Using Chebyshev's inequality, find an upper bound for $P(|X - 5| \geq 3)$.

b) Using Chebyshev's inequality, find a value k such that $P(|X - 5| \geq k) \leq 0.1$.

Exercise 1.2 [Easy]

Let $X_1, \ldots, X_n$ be i.i.d. random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu$. Using Hoeffding's inequality, find the smallest n such that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq 0.05\right) \leq 0.02.$$

$P\left(\frac{1}{n}\sum(X_i - E[X_i]) > t\right) \leq e^{-\frac{2n^2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}}$

solve for t $|\bar{X}_n - \mu| \leq t$

$\mu \in \left[\bar{X}_n - (b-a)\sqrt{\frac{\ln(2/\delta)}{2n}},\ \bar{X}_n + (b-a)\sqrt{\frac{\ln(2/\delta)}{2n}}\right]$

Take h(x) = e^sx, and X = $\frac{1}{n}\sum_{i=1}^{n}(X_i - E[X_i])$ mult n conviniently, apply Markov + hoeff lemma, then minimize over s and so:

Exercise 1.3 [Medium]

it bounds the sampled mean from the real mean difference

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ be a d-dimensional Gaussian random vector and let $Z = \|X\|^2$.

a) Compute $\mathbb{E}[Z]$ and $\mathrm{Var}(Z)$.

b) Using Chebyshev's inequality, show that

$\mathrm{Var}\left(\sum_{i=1}^{d} X_i^2\right) = \sum_{i=1}^{d} \mathrm{Var}(X_i^2)$ · because independent

$\lim_{n\to\infty} \mathbb{E}(|X_n - X|^p) = 0.$

$$P\left(\left|\|X\|^2 - d\sigma^2\right| > t\sqrt{d}\,\sigma^2\right) \leq \frac{2}{t^2}.$$

with expected value we care about magnitude

also, with Xn-X >= e is just yes or no. The p lets us set the sensibility.

c) What does this tell us about the distribution of $\|X\|$ as $d \to \infty$? Express your answer in terms of where the mass of X concentrates.

$\mathrm{P}(\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)) = 1.$

seq. we defined with rand. var. inside: x/n

$\lim_{n\to\infty} \mathrm{P}(|X_n - X| \geq \varepsilon) = 0.$

if seq. conv. dist. X with more than e prob 0

Exercise 1.4 [Medium]

Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli random variables with $P(X_i = 1) = p$ and $P(X_i = 0) = 1 - p$, and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

a) Derive upper bounds on $P(|\bar{X}_n - p| \geq \varepsilon)$ using both Chebyshev's and Hoeffding's inequality.

b) For $\varepsilon = 0.1$ and $p = 0.5$, determine numerically for which values of n the Hoeffding bound is tighter than the Chebyshev bound.

Exercise 1.5 [Hard]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ and define $Z = \|X\|^2$.

a) Show that $\mathbb{E}[\langle X, AX\rangle] = \sigma^2 \mathrm{tr}(A)$ for any symmetric matrix $A \in \mathbb{R}^{d\times d}$.

b) Let $B \in \mathbb{R}^{m\times d}$. Using part (a), show that $\mathbb{E}[\|BX\|^2] = \sigma^2\|B\|_F^2$.

c) Using Chebyshev's inequality and part (b), derive a bound on

$X^T B^T B X \neq X^T \|B\|_F^2 X$

you need to expand, take the tr and then convert to norm

watch out, take sum for tr, not var

$$P\left(\left|\frac{1}{d}\|BX\|^2 - \frac{\sigma^2\|B\|_F^2}{d}\right| > \varepsilon\right)$$

in terms of d, σ, $\|B\|_F$, $\|B\|_{\mathrm{op}}$ and ε. You may use $\mathrm{Var}(\|BX\|^2) \leq 2\sigma^4\|B\|_F^2\|B\|_{\mathrm{op}}^2$ without proof.

LLN = mean estimator converges when n large. SLLN -> bound on the deviation from the real mean

$$\mathrm{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X]\right| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(X)}{n\varepsilon^2}$$

$P\left(|\bar{X}_n - \mu| \geq \sigma\sqrt{\frac{\alpha}{n}}\right) \leq \frac{\sigma^2}{n \cdot \sigma^2 \frac{\alpha}{n}} = \frac{\sigma^2}{\sigma^2\alpha} = \frac{1}{\alpha}$

As n converges to infinity then the estimator mean converges to the ral mean. Proof: take var of the estimator mean, apply chebyshev, pick epsilon smartly so we get 1/alfa -> we kknow that with that prob the diff between est. and real will be

How many standard deviations away is the estimator mean from the real mean

CLT says it follows N(0, 1):

$$Z_n = \frac{\sqrt{n}}{\sigma}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right)$$

Z is the distribution of fluctuation of the mean when n -> infinity. Since we have the 1/n, taking var gives: n*2/n² so we cancelwith that

Law of Large Numbers & Central Limit Theorem

Exercise 2.1 [Easy]

Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_i] = 3$ and $\mathrm{Var}(X_i) = 12$.

a) What does the Law of Large Numbers tell us about $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ as $n \to \infty$?

b) Using the Central Limit Theorem, state the approximate distribution of $\bar{X}_n$ for large n.

c) Approximate $P(\bar{X}_{100} > 3.5)$ using the CLT. You may use $\Phi(1.44) \approx 0.925$, where Φ is the standard normal CDF.

Exercise 2.2 [Medium]

A biased die has probability distribution: $P(X = k) = \frac{k}{21}$ for $k = 1, 2, \ldots, 6$.

a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

b) If the die is rolled $n = 200$ times, use the CLT to approximate the probability that the sample mean exceeds 4.5. You may use $\Phi(1.61) \approx 0.946$. This means P(Z<=1.61) = 0.94

c) Using Chebyshev's inequality, find the minimum number of rolls n needed to guarantee that $P(|\bar{X}_n - \mathbb{E}[X]| \geq 0.5) \leq 0.05$.

Therefore compute Z with X = 4.5, and that is the p. of P(z>= 1.61)

Which means the p. of being 1.61 std. deviation bigger

Exercise 2.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define $S_n = \sum_{i=1}^{n} X_i$.

a) Show using Chebyshev's inequality that $\bar{X}_n = \frac{S_n}{n}$ converges to μ in probability (this is the Weak Law of Large Numbers). recall the /n in v

b) Let $g : \mathbb{R} \to \mathbb{R}$ be a continuous function. Using the CLT, show that for large n:

$$\mathbb{E}[g(\bar{X}_n)] \approx g(\mu) + \frac{\sigma^2}{2n} g''(\mu).$$

Hint: Use a second-order Taylor expansion of g around μ and the CLT approximation.

c) Apply part (b) to estimate $\mathbb{E}[e^{\bar{X}_n}]$ when $\mu = 0$ and $\sigma^2 = 1$, and compare it to the exact value e^{μ} as $n \to \infty$.

Convergence of Random Variables

Exercise 3.1 [Easy]

State whether each of the following implications is true or false. Give a brief justification.

a) Convergence in probability implies convergence in distribution.

b) Convergence in distribution implies convergence in probability.

c) If $X_n \xrightarrow{P} c$ for a constant c, does $X_n \xrightarrow{d} c$?

Convergence distributions: FXn (x)!'FX (x) at every point ->

convergence in probability: P("#Xn " X"#>μ)!'0 for all μ

FXn(x) -> F(-Xn)(x) converges in distributions (same), but i

Exercise 3.2 [Medium]

Let $U_1, U_2, \ldots$ be i.i.d. Uniform$(0, 1)$ random variables and define $M_n = \max(U_1, \ldots, U_n)$.

a) Show that $P(M_n \leq x) = x^n$ for $x \in [0, 1]$.

b) Determine whether M_n converges in probability to some limit. If so, identify the limit.

c) Find the distribution of $Y_n = n(1 - M_n)$ as $n \to \infty$. What type of convergence does this represent?

Exercise 3.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. Uniform$(0, 1)$ and define

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \frac{1}{3}\right).$$

a) Compute $\mathbb{E}[X_i^2]$ and $\mathrm{Var}(X_i^2)$.
b) Using the CLT, determine the asymptotic distribution of Y_n.
c) Use part (b) to approximate $P(Y_{400} > 0.05)$. You may use $\Phi(0.47) \approx 0.681$.
d) Is it possible to apply Hoeffding's inequality to bound $P\big(\big|\frac{1}{n}\sum_{i=1}^n X_i^2 - \frac{1}{3}\big| \geq t\big)$? If so, state the bound.

Confidence Intervals for the Test Error

Exercise 4.1 [Easy]

A classifier f is evaluated on $n = 1000$ i.i.d. test samples and achieves an empirical test error of $\hat{\mathrm{err}}_f = 0.08$.

a) Using Hoeffding's inequality, construct a 95% confidence interval for the true error err_f.
b) How many test samples are needed to guarantee the confidence interval has width at most 0.02 at 95% confidence?

Exercise 4.2 [Medium]

Let f be a binary classifier with true error $\mathrm{err}_f = P(f(X) \neq Y)$, evaluated on n i.i.d. test samples.

a) Write $\hat{\mathrm{err}}_f$ in terms of the indicator variables $\mathbf{1}_{f(X_i)\neq Y_i}$ and show that $\mathrm{Var}(\hat{\mathrm{err}}_f) = \frac{\mathrm{err}_f(1-\mathrm{err}_f)}{n}$.
b) Using the fact that $\mathrm{err}_f(1 - \mathrm{err}_f) \leq \frac{1}{4}$, derive a confidence interval via Chebyshev's inequality.
c) For $n = 10{,}000$ and $\varepsilon = 0.01$, compare the upper bounds from Chebyshev and Hoeffding on $P(|\hat{\mathrm{err}}_f - \mathrm{err}_f| \geq \varepsilon)$. Which one is tighter?

Exercise 4.3 [Hard]

You are evaluating two classifiers f and g on the same test set of n i.i.d. samples. Define $\Delta = \mathrm{err}_f - \mathrm{err}_g$ and $\hat{\Delta} = \hat{\mathrm{err}}_f - \hat{\mathrm{err}}_g$.

a) Let $Z_i = \mathbf{1}_{f(X_i)\neq Y_i} - \mathbf{1}_{g(X_i)\neq Y_i}$. Express $\hat{\Delta}$ in terms of Z_i and note that $Z_i \in [-1, 1]$.
b) Using Hoeffding's inequality on Z_i, derive a confidence interval for Δ.
c) If $n = 5000$, $\hat{\mathrm{err}}_f = 0.12$ and $\hat{\mathrm{err}}_g = 0.10$, determine with 95% confidence whether there is a statistically significant difference between the two classifiers.

Mixed Challenging Exercises

Exercise 5.1 [Hard]

A machine learning engineer trains a model and wants to evaluate it rigorously. The test set has n i.i.d. samples, each in $\mathbb{R}^d$, with labels in $\{0, 1\}$.

a) The model achieves $\hat{\mathrm{err}}_f = 0.03$ on $n = 500$ test samples. Using Hoeffding's inequality, compute a 99% confidence interval for err_f.
b) The engineer claims: "My model has at most 5% error." Using part (a), can this claim be supported at 99% confidence?
c) How large must n be so that a 99% confidence interval using Hoeffding has half-width at most 0.01?
d) Using the CLT, approximate the distribution of $\hat{\mathrm{err}}_f$ for $n = 500$ assuming $\mathrm{err}_f = 0.03$. Compare the CLT-based 99% confidence interval to the one from Hoeffding. You may use $z_{0.005} = 2.576$.

Exercise 5.2 [Hard]

Consider a dataset of n i.i.d. samples $X_1, \ldots, X_n$ drawn uniformly from $[0,1]^d$.

a) For a fixed point $p \in [0,1]^d$, compute $\mathbb{E}[\|X_i - p\|^2]$ and show that it does not depend on p when $p = (\frac{1}{2}, \ldots, \frac{1}{2})$. *Hint:* Use $\mathrm{Var}(X_{i,j}) = \frac{1}{12}$ for $X_{i,j} \sim \mathrm{Uniform}(0,1)$.

b) Using Hoeffding's inequality applied coordinate-wise, show that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_{i,j} - \frac{1}{2}\right| \geq t\right) \leq 2e^{-2nt^2}$$

for each coordinate j.

c) Using a union bound and part (b), find a value t such that with probability $\geq 1-\delta$, the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ satisfies $\|\bar{X}_n - (\frac{1}{2}, \ldots, \frac{1}{2})\|_\infty \leq t$.

d) Interpret the result: how does the number of samples n needed to achieve $\|\bar{X}_n - \mu\|_\infty \leq \varepsilon$ with confidence $1-\delta$ scale with the dimension d?